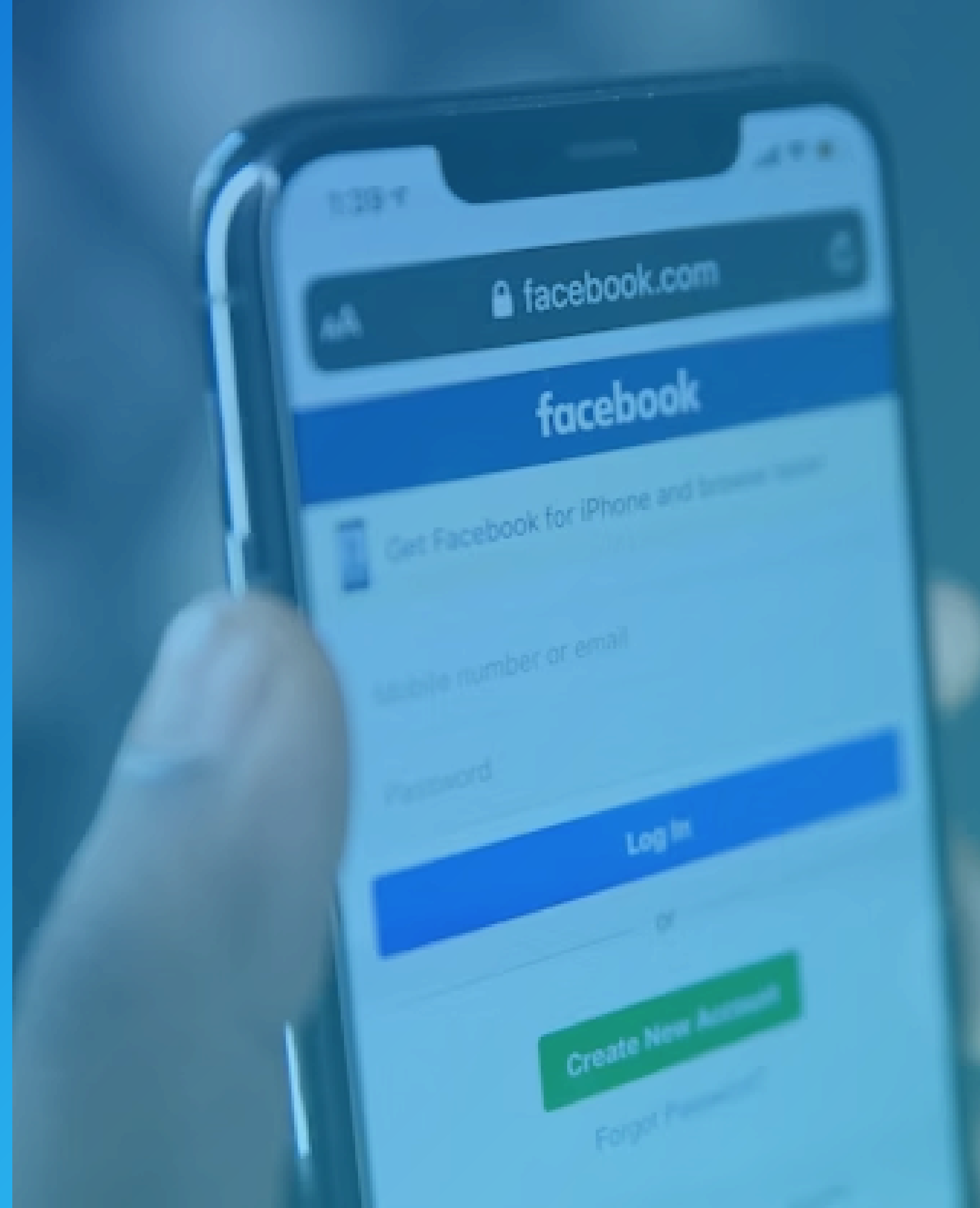


Data Analyst Project

Facebook Content Monetization Revenue Prediction and Analysis

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July 2026



Understanding the business problem, objectives, and analytical framework.

Business Understanding

As competitors' Facebook monetization revenue is not publicly disclosed, machine learning models are developed to estimate revenue using available performance metrics. These estimates competitive analysis and help identify opportunities to improve Pikiran Rakyat's Facebook monetization strategy.

Objectives

- 01
- Predict competitors' Facebook monetization revenue using machine learning models based on publicly available performance metrics.
- 02
- Identify the key performance metrics that drive Facebook monetization revenue.
- 03
- Analyze revenue patterns across media profiles and over time to uncover monetization strategies.

Framework

Data Understanding	Data Preparation	Modelling	Evaluation	EDA	Deployment
Obtain and explore Facebook data from Meta Business Suite and SocialInsider.	Process, transform, clean, standardize, and encoded the raw dataset into an modelling-ready format.	Split 80%/20% dataset and train Linear Regression, Random Forest, and XGBoost.	Compare R ² , MAE, and RMSE.	Identify key performance metrics that drive revenue and revenue patterns across media profiles.	Recommendations for Facebook monetization strategy.

Dataset Overview

Understanding dataset structure, quality, and workflow.

Dataset Overview

4,020

Total Posts

53

Total Variables

8

Media Profiles

2

Data Sources

Overview	Pikiran Rakyat	Competitors
Period	1 May–7 June	1–7 June
Rows	763	3,257
Columns	35	18
Missing Values	5,901	5,799
Duplicate Rows	0	0
Source	Meta Business Suite	SocialInsider

The EDA will be conducted using data from the first week of June only. However, the May dataset from Pikiran Rakyat was required for model training to predict competitors' revenue.

Dataset Pre-Processing

Pikiran Rakyat

763 rows
35 columns

1. Handling Missing Values

- Filled missing values using corresponding English columns (e.g. ID Halaman ← Page ID, etc.)
- Removed the English columns after filling

2. Feature Selection

- Dropped 19 unnecessary columns (metadata, technical metrics, etc.)
- Kept 11 relevant columns for analysis

3. Output

- Obtained cleaned Pikiran Rakyat's Facebook data with:
 - 763 rows
 - 11 columns

Competitors

3,257 rows
18 columns

1. Handling Missing Values

- Filled missing Impressions with Video Views
- Removed redundant column (Video Views)
- Removed missing Reach values

2. Feature Selection

- Dropped 6 unnecessary columns (platform, engagement rate, followers, pillars, country)
- Kept 12 relevant columns for analysis

3. Output

- Obtained cleaned Competitors' Facebook data with:
 - 2,521 rows
 - 12 columns

4. Data Integration

- Unified column names across datasets
- Aligned metrics (e.g. Likes → Reactions)
- Combine into one master data

5. Data Standardization

- Standardize post type (Photo, Video, Link, Text)
- Standardize date format to MM/DD/YYYY HH:mm

6. Final Output

- Obtained cleaned master data with:
 - 3,284 rows
 - 12 columns

Project Overview

Dataset Overview

Modelling

Analysis

Recommendations

Closing

02

Preparing the dataset and features to predict missing Revenue values for competitor posts.

Why Revenue Prediction?



2,521 competitor posts have missing Revenue values.



Revenue estimation was required to enable competitor benchmarking and analysis.

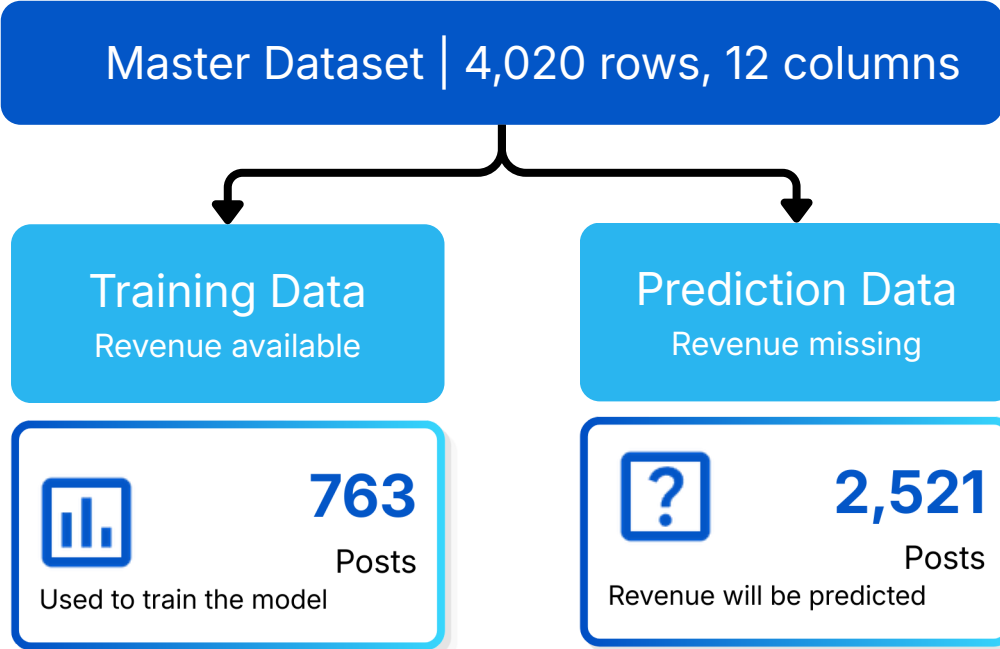


Missing revenue values were predicted using machine learning before analysis.

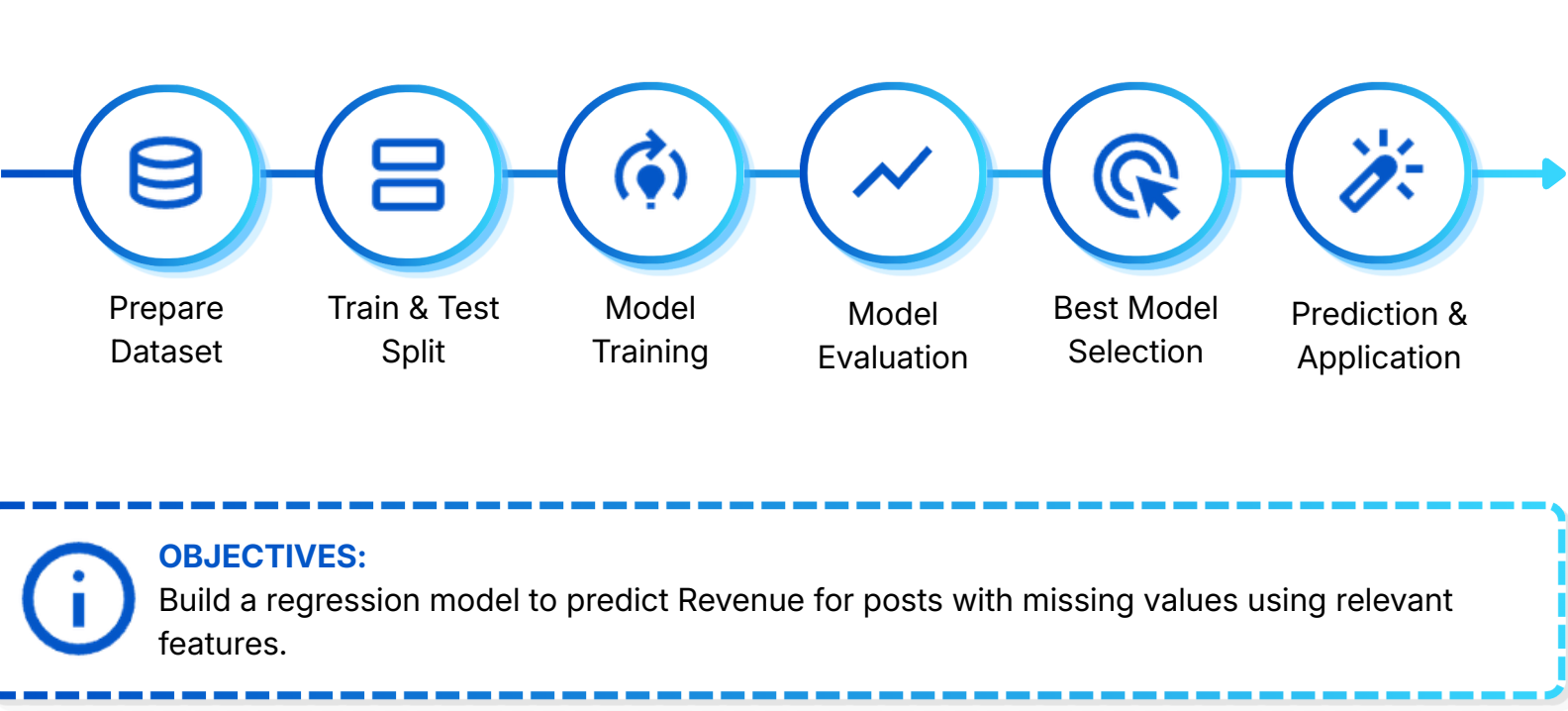


TARGET VARIABLE: Revenue (Log-transformed)

Data Split for Reach Prediction



Modelling Workflow



Feature Engineering



PROFILE INFORMATION

- Encoded Profile (e.g. 0: Antaranews , 1: Detik, 2: Kompas, etc.)

ENCODING



CONTENT FEATURES

- Encoded Post Type (e.g. 0: Link, 1: Photo, etc.)
- Word Count
- Hashtag Count

DERIVED + ENCODING



POSTING TIME FEATURES

- Posting Hour
- Encoded Day of the Week (e.g. 0: Friday, 1: Monday, etc.)

DERIVED + ENCODING



METRICS FEATURES

- Engagement
- Impressions
- Reach
- Reactions
- Comments
- Shares

LOG TRANSFORMATION

Train & Test Split

Train Set (80%)

610 Posts
Used to train the model

Test Set (20%)

153 Posts
Used to evaluate the model

RANDOM STATE: 42

Output

2,521 posts of Competitors were filled with predicted Revenue values.

Modelling Approach

Comparing Linear Regression, Random Forest, and XGBoost for Revenue prediction.

Modelling Workflow

**1. Prepare Dataset**
Cleaned dataset with engineered features and encoded variables.

**2. Train & Test Split**
Split data into training (80%) and testing (20%).

**3. Model Training**
Train multiple models using the training dataset.

**4. Model Evaluation**
Evaluate models on test set using multiple metrics.

**5. Best Model Selection**
Select the best performing model for prediction.

**6. Prediction & Application**
Generate predictions for missing values and downstream analysis

Models Evaluated

**Linear Regression**

A linear model that assumes a linear relationship between features and target variable.

 **STRENGTHS**

- Simple, fast, and easy to interpret
- Works well when relationships are linear
- Provides clear insights into feature impact

 **LIMITATIONS**

- Assumes linearity between features & target
- Sensitive to outliers
- May underperform on complex patterns

**Random Forest**

An ensemble of decision trees using bagging technique to improve accuracy and reduce overfitting.

 **STRENGTHS**

- Captures non-linear relationships
- Robust to outliers and noise
- Provides feature importance

 **LIMITATIONS**

- Less interpretable than linear models
- Can be computationally intensive
- Performance depends on parameter tuning

**XGBoost**

A gradient boosting algorithm that builds trees sequentially to correct errors from previous tree.

 **STRENGTHS**

- High predictive performance
- Handles complex patterns effectively
- Built-in regularization to reduce overfitting

 **LIMITATIONS**

- More sensitive to hyperparameters
- Requires more computational resources
- Less interpretable than linear models

Evaluation Metrics

R²

Measures the proportion of the variance explained by the model (0–1).

Higher is better

Root Mean Squared Error (RMSE)

Measures the average magnitude of prediction errors. Penalizes large errors more.

Lower is better

Mean Absolute Error (MAE)

Measures the average absolute difference between actual and predicted values.

Lower is better

Prediction Results & Comparison

XGBoost was selected as the best model for predicting Revenue (Log-transformed)

Evaluation Metrics (Test Set)

R² Score

0.7018

Higher is better

RMSE

0.2744

Lower is better

MAE

0.0728

Lower is better

Model Comparison (Test Set)

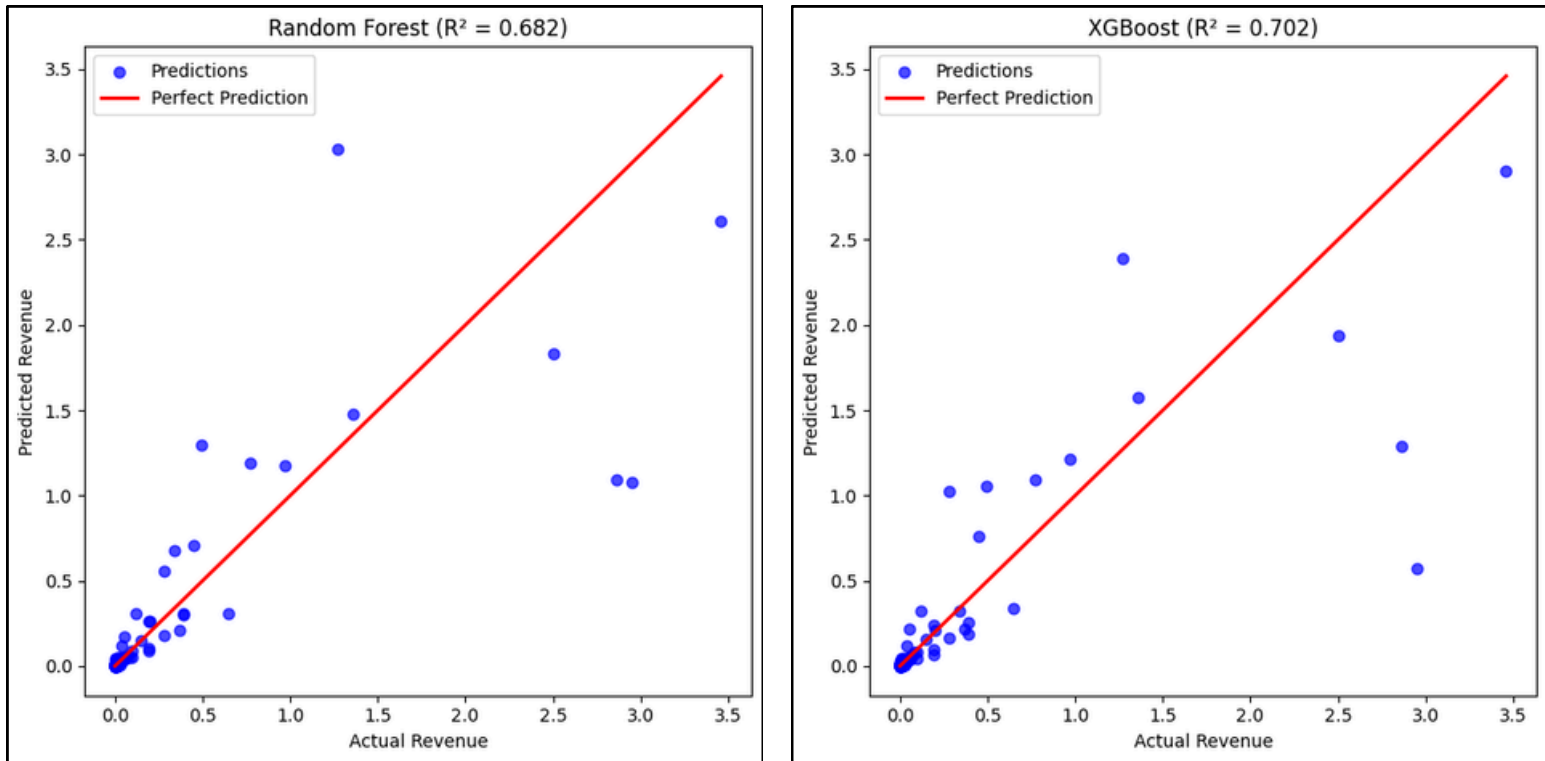
Model	MAE	RMSE	R ²
Linear Regression	0.181161	0.385890	0.409985
Random Forest	0.076570	0.283328	0.681936
XGBoost	0.072836	0.274354	0.701766

BEST MODEL

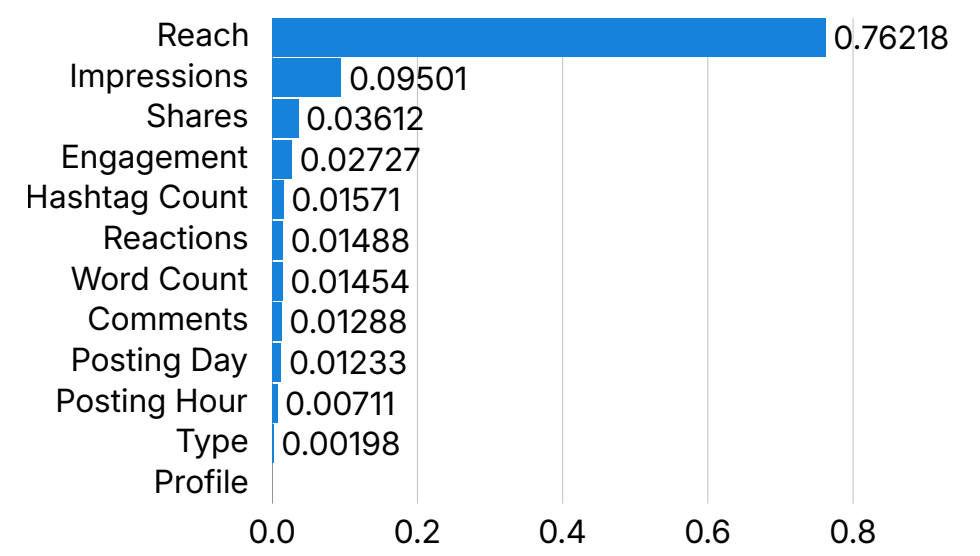
XGBoost

MAE 0.0728
RMSE0.2744
R² 0.7018

Actual vs Predict of Two Best Models



Feature Importance



Feature Importance Explanation

Reach (Log)
The most important feature, contributing 76.22%. Revenue was strongly driven by the amount of reach achieved by a post.

Impressions (Log)
Higher impressions increased the opportunity to generate monetization.

Shares and Engagement (Log)
Shares helped expand distribution and may lead to higher monetization. Engagement reflect audience interactions can improve content visibility and monetization potential.

Content & Temporal Feature
Word Count, Hashtag Count, Posting Day, Hour, and Content Type also contributed, but to a smaller extent.

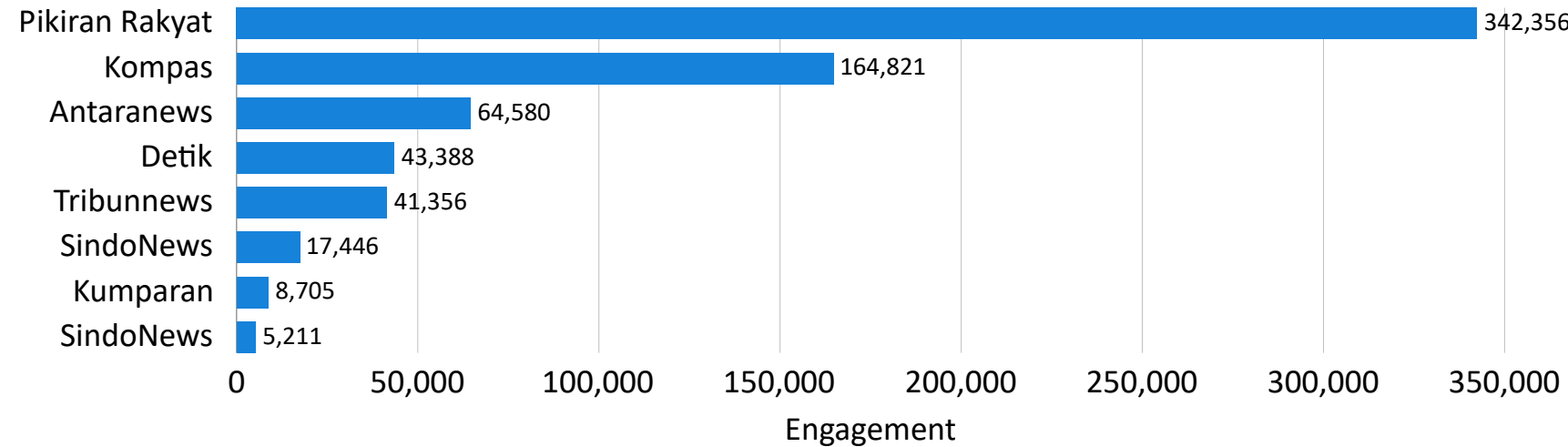
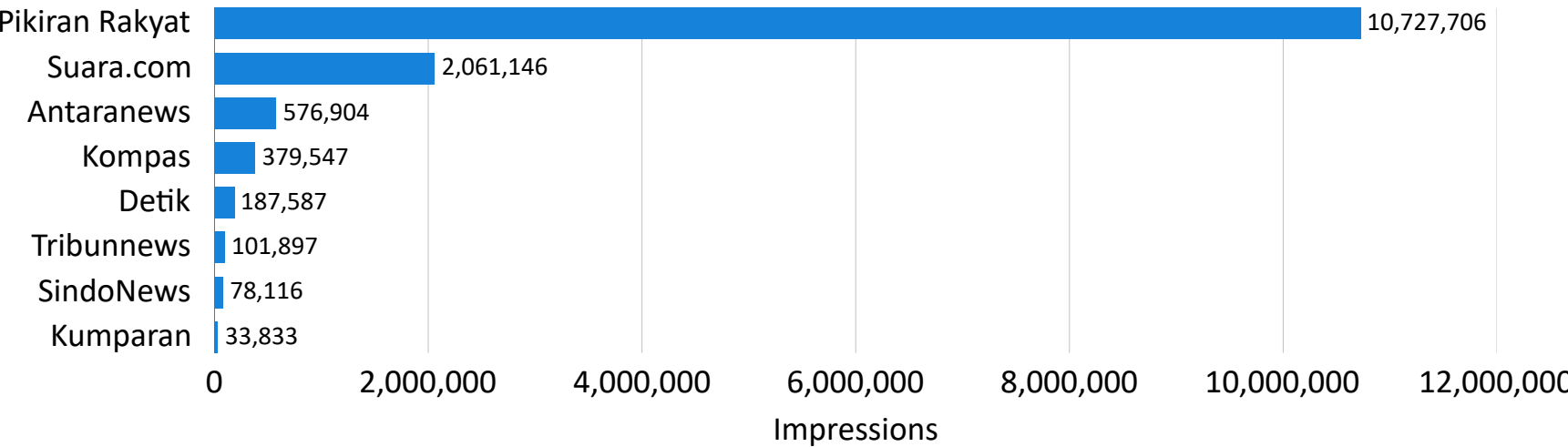
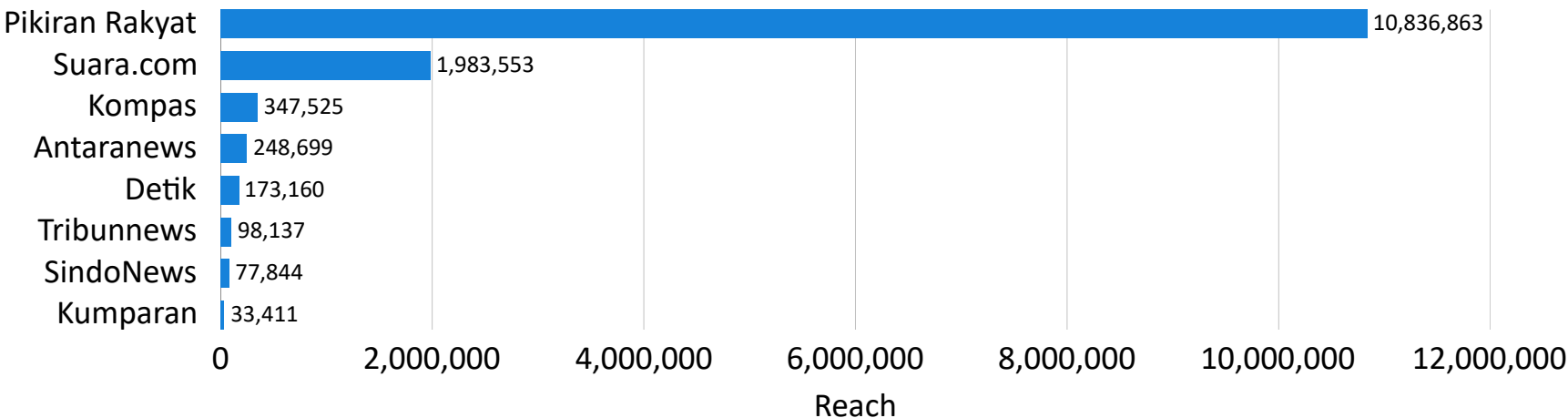
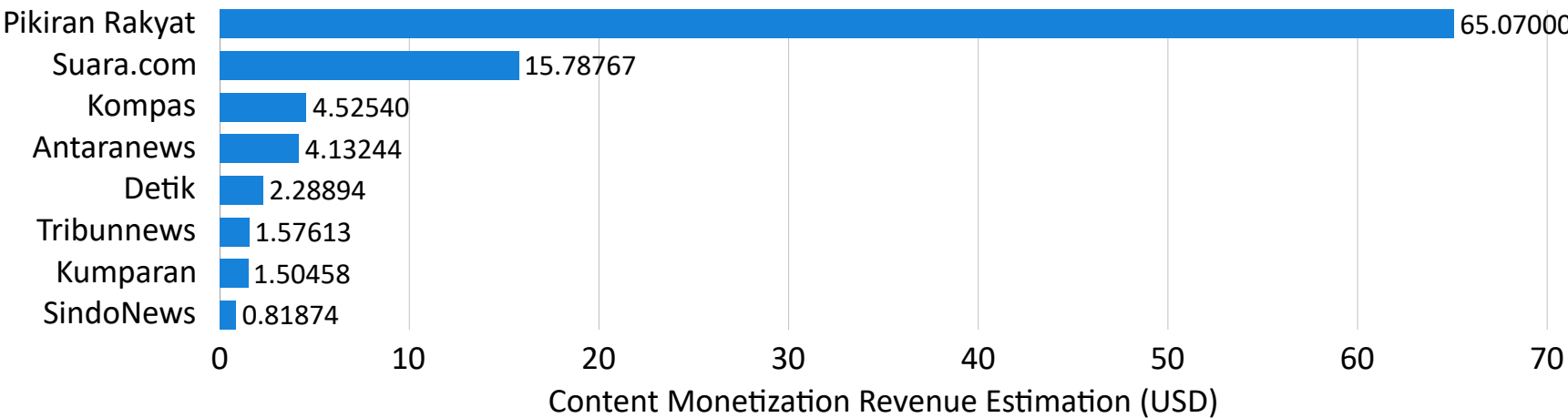
Key Insights

- XGBoost achieved the best performance across all metrics with R² of 0.7018, indicating that 70.18% of the variance in revenue was explained by the model.
- The model showed good predictive ability with low MAE (0.0728) and RMSE (0.2744) on the test set.
- Reach (log) was the most dominant feature, emphasizing the wider reach drives higher revenue, followed by Impressions (log) and engagement-related metrics (log).
- The model was reliable and was used to predict missing Revenue values for competitors.

XGBoost was selected as the final model for Revenue prediction.

Prediction Results & Comparison

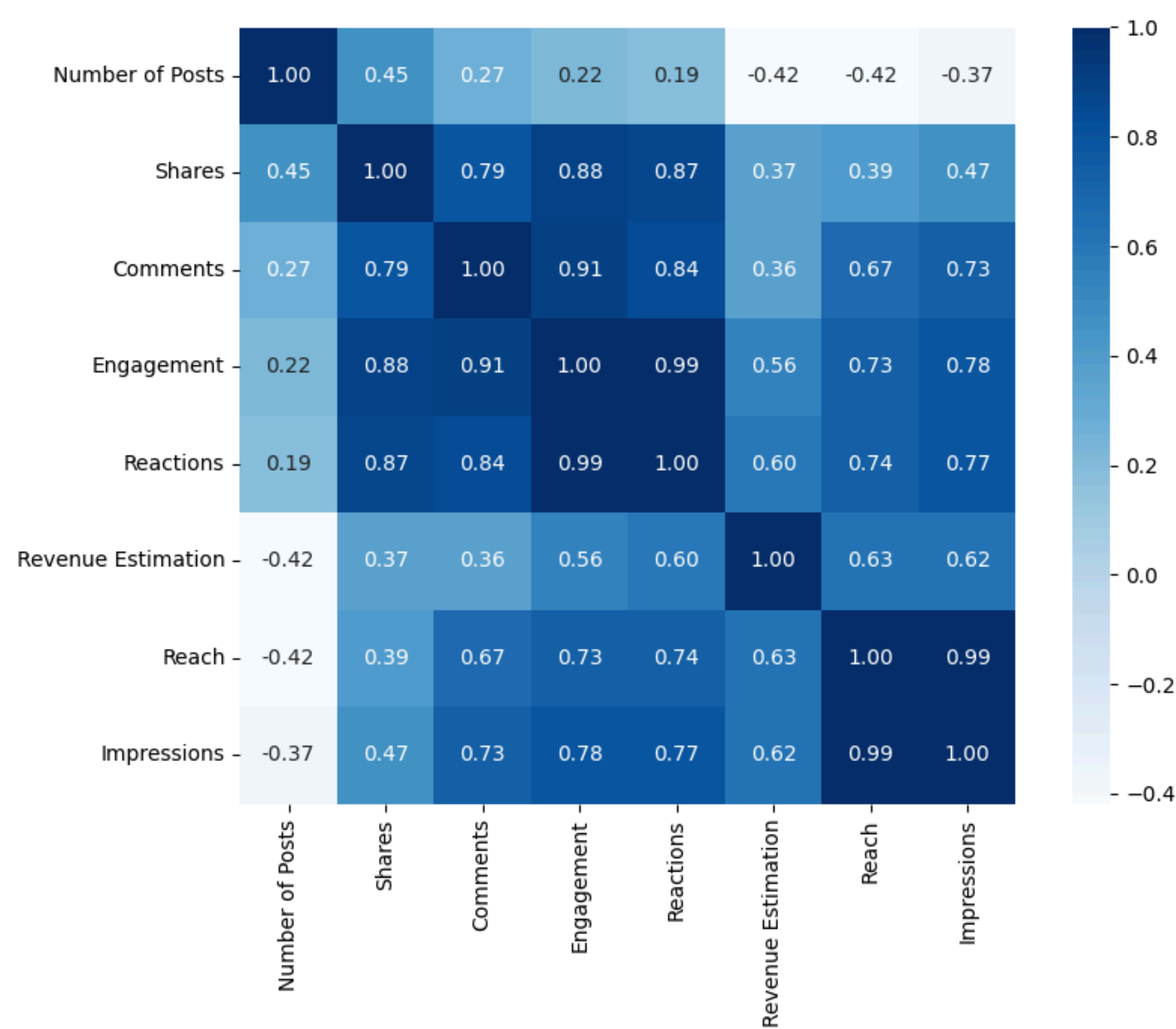
Revenue was driven by distribution scale rather than engagement alone.



Key Insights

- ✓ Pikiran Rakyat generated the highest estimated revenue, substantially outperforming all competitors.
- ✓ Pikiran Rakyat also recorded the highest Reach and Impressions, indicating the broadest content distribution.
- ✓ Suara.com achieved the highest Engagement, but its estimated revenue remained considerably lower than Pikiran Rakyat.
- ✓ The results suggested that audience engagement alone did not directly translate into higher revenue.

Revenue was driven more by content distribution than posting volume.



Key Insights

- Revenue showed stronger correlations with Reach (0.63) and Impressions (0.62) than engagement metrics.
- Number of Posts had a negative relationship with revenue, indicating higher posting volume did not guarantee better monetization.

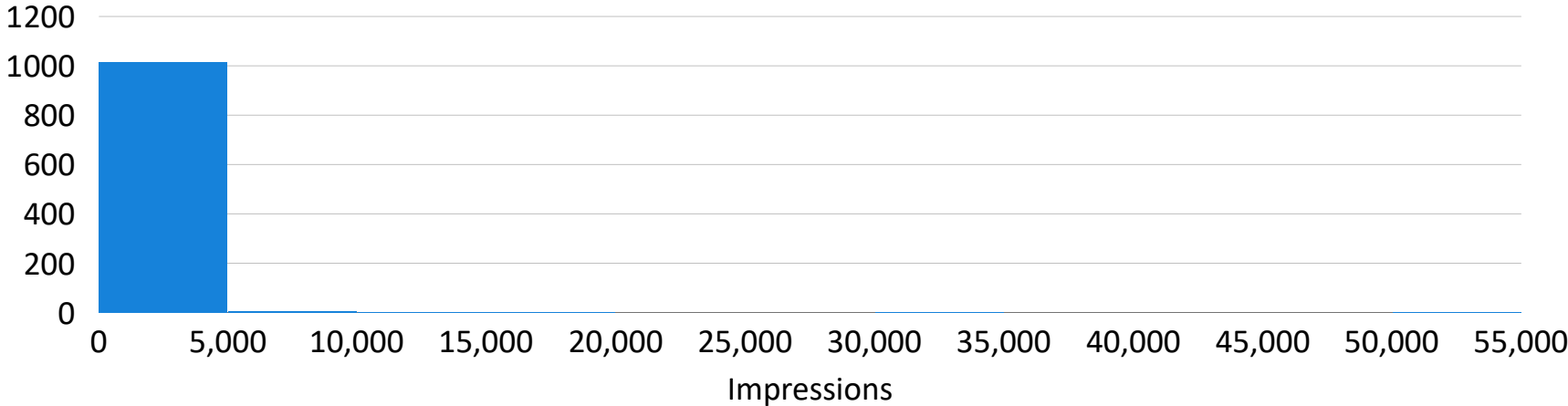
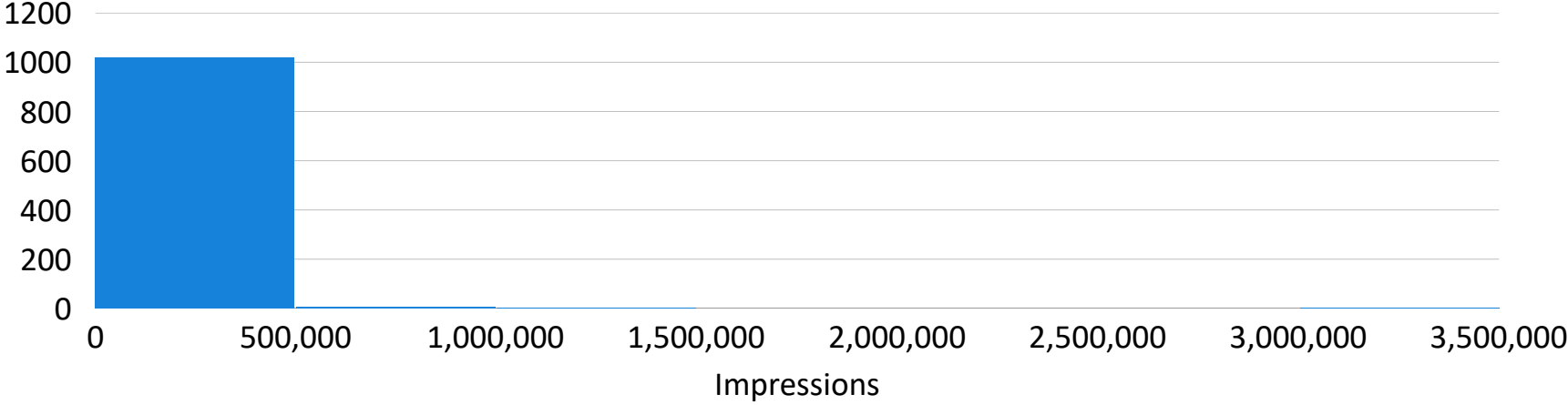
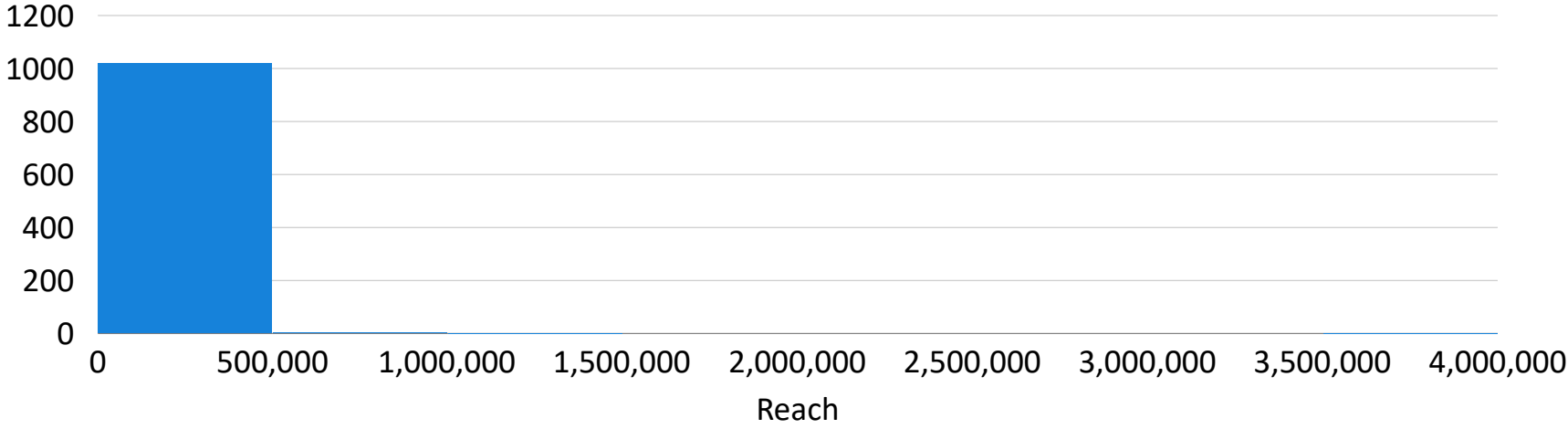
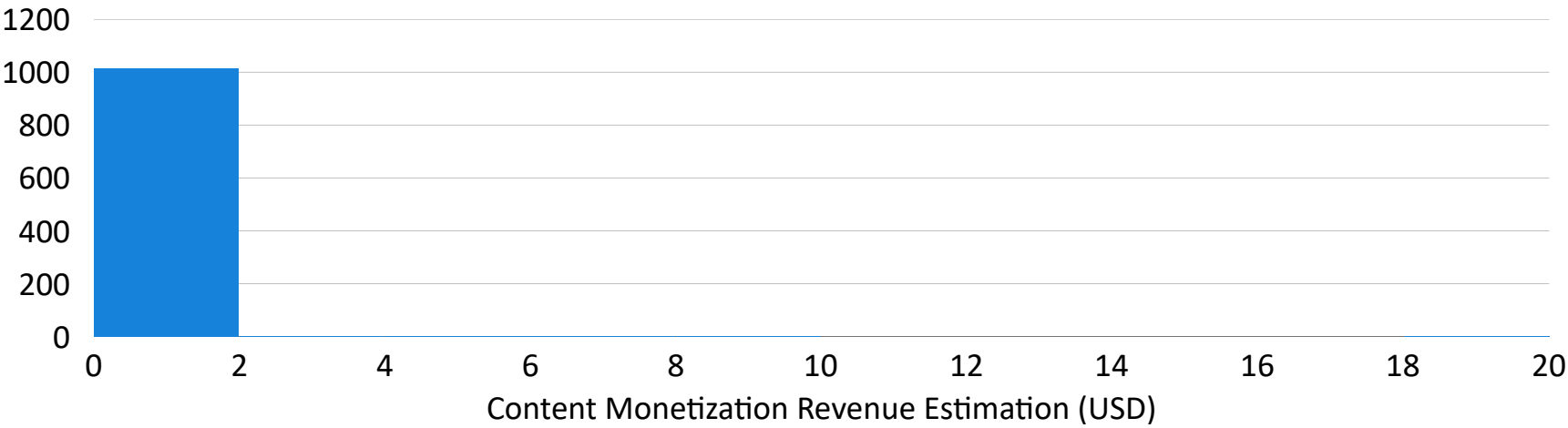
Key Takeaway

Content distribution scale was a stronger monetization driver than posting frequency or engagement alone.

Key Impact

Focus on expanding content reach and impressions to maximize revenue potential.

Right-skewed performance distribution requires median-based evaluation to reduce outlier impact.

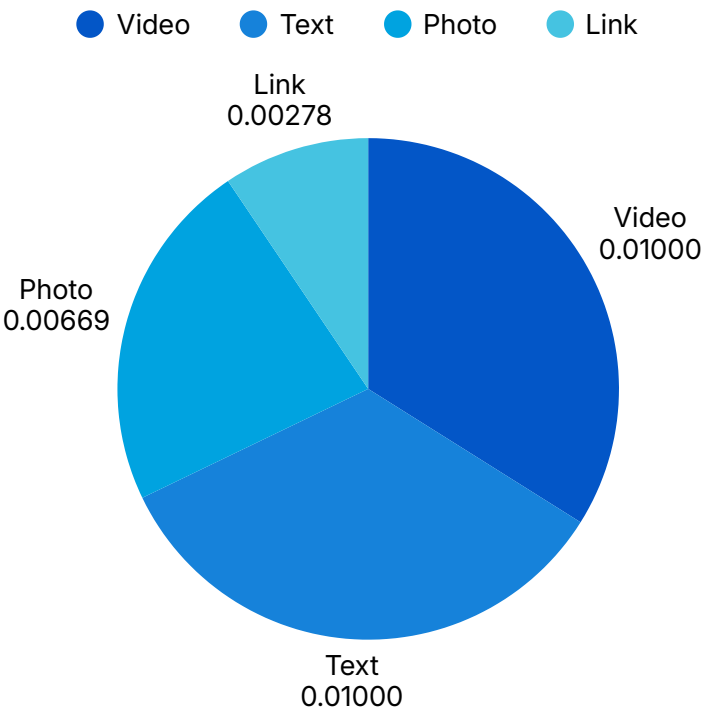


Key Insights

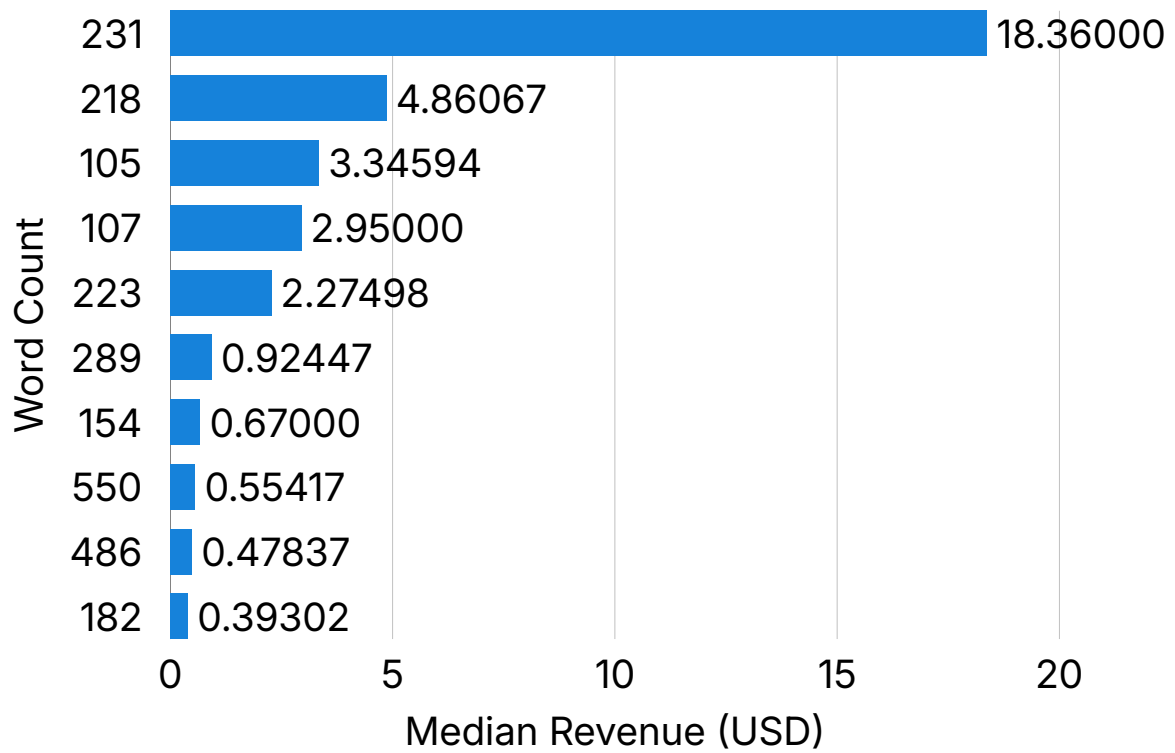
- ✓ Revenue, Reach, Impressions, and Engagement exhibit positively skewed distributions.
- ✓ Most posts have relatively low values, while a small number of posts generate exceptionally high results (long-tail pattern)
- ✓ The presence of extreme outliers indicates that the mean can be heavily influenced by a few high-performing posts.
- ✓ To ensure more robust and representative analysis, median values will be used for further analysis.

Content format and caption strategy influenced monetization performance.

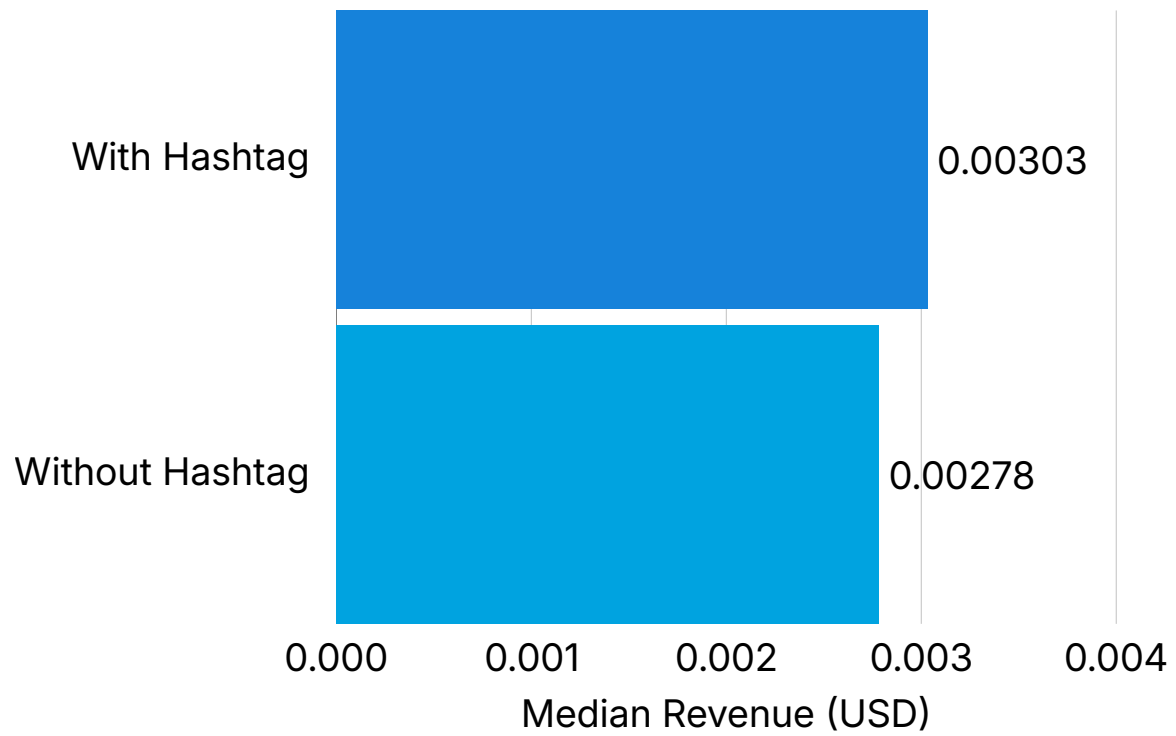
Median Revenue by Content Type



Median Revenue by Caption Length



Median Revenue by Hashtag Presence



Key Insights

- ✔ Video and Text deliver the highest median revenue, significantly outperforming Photos and Links.
- ✔ Captions with around 231 words achieved the highest median revenue, while very short or very long captions perform less.
- ✔ Using hashtags provides a small, but positive uplift in median revenue.
- ✔ Content format, caption length, and hashtag usage also had roles in monetization performance, though to a smaller extent.

Key Takeaway

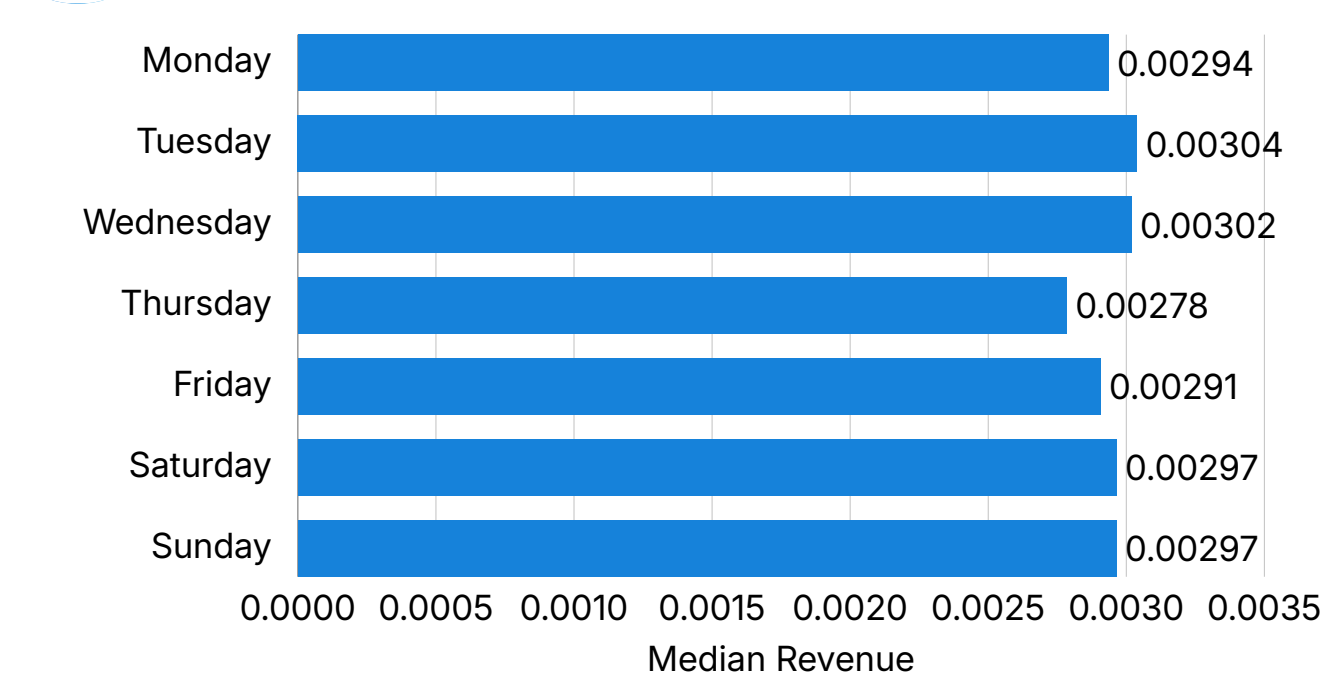
Content structure and optimization factors supported monetization, but were secondary to distribution scale.

Key Impact

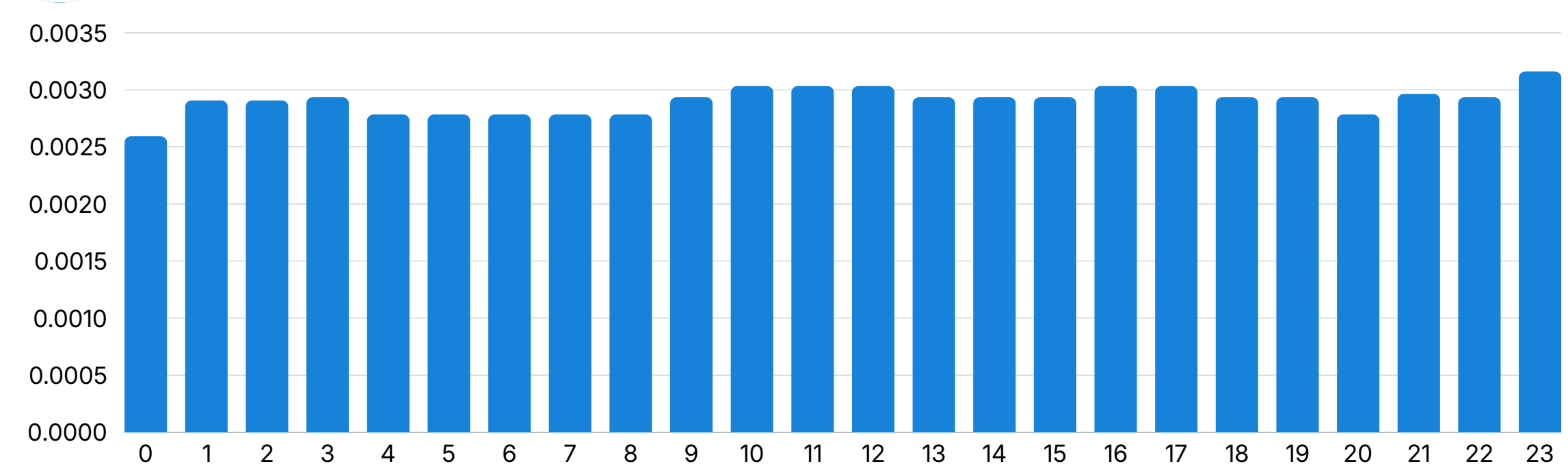
Optimize content formats, caption length, and hashtag usage to improve revenue potential.

Posting timing had limited impact on revenue compared to content factors.

Median Revenue by Posting Day



Median Revenue by Posting Hour



Key Insights

- Posting performance was relatively consistent across days of the week.
- Tuesday tended to yield the highest median revenue, while Thursday performed the lowest.
- Revenue remains stable across most posting hours, with the best range for posting was 10 AM–12 PM.
- While timing has some impact, content quality and other factors likely play a larger role in revenue.

Key Takeaway

Content quality and distribution factors were more influential than posting timing.

Key Impact

Prioritize content optimization over adjusting posting schedules to improve revenue performance.

Highlighting the top 3 posts that generated the highest estimated revenue.

Rank	Preview	Content Overview	Post info	Engagement Metrics	Reach & Impressions	Estimated Revenue (USD)
1		"Jeddah Fashion Week" Ala Jemaah Haji Makassar Lenggak-lenggok parade busana jemaah haji perempuan asal Makassar Sulawesi Selatan mencuri perhatian di Bandara Internasional King Abdulaziz Jeddah Video	6 June 2026 Saturday 00:33 231 words 0	Reactions 4,010 Comments 2,107 Shares 423 Engagement 6,540	Impressions 747,366 Reach 755,922	18.36 \$
2		Tinjau MBG di SMPN 111 Jakarta Barat, Prabowo: Makannya Enak Gak? Presiden Prabowo Subianto meninjau pelaksanaan Program Makan Gratis (MBG) di SMP Negeri 111 Jakarta Barat, Selasa, 2 Juni 2026. Video	2 June 2026 Tuesday 05:36 218 words 4	Reactions 40,864 Comments 12,032 Shares 735 Engagement 53,631	Impressions 3,273,767 Reach 3,509,005	9.70 \$
3		Prabowo Mengaku Sedih Usai Rombak Pimpinan BGN Presiden Prabowo Subianto, mengungkapkan rasa sedihnya saat harus mengambil keputusan sulit untuk mengganti jajaran pimpinan di Badan Gizi Nasional (BGN). Video	3 June 2026 Wednesday 08:53 105 words 4	Reactions 8,809 Comments 6,820 Shares 394 Engagement 16,023	Impressions 537,734 Reach 557,736	6.69 \$

Key Insights

All top 3 performing posts were Videos, confirming that this format drives higher monetization potential.

High engagement combined with broad reach and impressions consistently leads to higher revenue.

News and Human Interest topics (Hajj stories, government programs, investigation) resonate strongly with audience.

All top 3 posts were from Pikiran Rakyat accounts, confirming the dominance in estimated revenue than competitors.

Business recommendations, actions, and expected impacts.

Recommendations	Key Finding	Recommended Action	Expected Impact
 Prioritize Video Content	Video content generated the highest median revenue and dominated the top performing posts.	Increase the proportion of video content, especially on high interest topics, such as News and Human Interest.	Higher Revenue, Reach, Impressions, and Engagement.
 Strengthen Content Distribution	Reach and Impressions were the strongest drivers of revenue in the prediction model.	Improve content distribution through a better headlines, thumbnails, and promotions.	Broader audience exposure and higher revenue potential.
 Optimize Caption Strategy	Captions around 231 words and hashtag usage produced slightly higher median revenue.	Use informative captions with optimal length and relevant hashtags.	Better audience exposure and interaction also higher revenue potential.
 Prioritize Posting on Prime Day and Time	Posts published at Tuesday and 11:00 PM generated higher median revenue.	Prioritize posting at Tuesday and optimal hours, such as 11:00 PM or 10 AM–12 PM.	Higher revenue potential.
 Prioritize Content Quality over Quantity	Number of posts daily had negative correlation with revenue	Focus on producing content that can reach wider audiences and invite huge interactions rather than increasing posting frequency.	More effiecient resources allocation and sustainable monetization growth.

Summary of key takeaways, limitations, and suggested next steps.

Conclusion



Developed a machine learning model to estimate Facebook Content Monetization Revenue using content characteristics, posting information, and engagement metrics.



XGBoost achieved the best performance ($R^2 = 0.7018$, RMSE = 0.2744, and MAE = 0.0728) and was selected to estimate missing revenue values.



Reach, Impressions, and engagement-related metrics were identified as the primary drivers of monetization revenue, while content type, caption characteristics, and posting time provided additional, but smaller contributions.

Limitations



The model was trained using only one month historical data, limiting its ability to capture longer-term seasonal trends.



Several potentially important features contained substantial missing values or were unavailable, limiting the model's predictive performance.



The model estimated revenue based on observed post-performance metrics rather than forecasting revenue before content is published.

Next Steps



Collect additional historical data covering multiple months or years to improve model robustness.



Incorporate richer features such as demographics, duration, topic classification, clicks, sentiment, etc.



Perform hyperparameter optimization and evaluate additional regression models to improve prediction.

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